

- 1 **Trypanosoma cruzi**
 - Causative agent of Chagas disease
 - Discovered by Carlos Chagas
 - Named after scientist, Brazilian Guy
- 2 **Chagas disease**
 - Top 10 infectious diseases
 - 10th leading cause of death
 - 10-20% mortality
 - Leading cause of cardiac disease in South and Central America
- 3 **Vectors of Chagas disease**
 - Triatomine bugs
 - Reduviid bugs
 - Cimex bugs
 - Bed bugs
 - Lice
- 4 **Means of transmission**

Category	Comments
Direct	Transfusion bugs (contaminated blood)
Transfusion	Organ donor
Organic	Any stage of T. cruzi infection (transmission from animal, neonatal death)
Accidental	Ingestion of food contaminated with trypomastigotes (unpasteurized, laboratory accidents)

Chagas disease

=16-18 million infected

=100 million at risk

= 50,000 deaths annually

= leading cause of cardiac disease in South and Central America

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Vectors of Chagas disease

- Common Names
- triatomine bugs
 - reduviid bugs
 - assassin bugs
 - kissing bugs
 - cone nose bugs



- Genera
- Triatoma
 - Rhodnius
 - Panstrongylus

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Modes of transmission

Source	Comments
Vector	<u>Triatomine bugs</u> (contaminated feces)
Transfusion	Urban areas
Congenital	Any stage of <u>T. cruzi</u> infection [Premature labor, abortion, neonatal death]
Accidental	Ingestion of food contaminated with <u>metacyclic trypomastigotes</u> ; laboratory accidents

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Chagas disease: disease stages

Acute Phase

- active infection
- 1-4 months duration
- most are asymptomatic (children most likely to be symptomatic)

Indeterminate Phase

- 10-30 years of latency
- relatively asymptomatic with no detectable parasitemia
- seropositive

Chronic Phase

- 10-30% of infected exhibit cardiomyopathy or megasyndromes

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3 vectors of Chagas disease

4 Modes of transmission

5 Chagas disease: disease stages

6 Chagas disease: disease stages

Chagas disease: disease stages

Acute Phase features

- 1-2 week incubation period
- local inflammation
Romana's sign
Chagoma
- symptoms can include: fever, malaise, lymphadenopathy, hepatosplenomegaly, nausea, diarrhea
- acute, often fatal, myocarditis develops in a few individuals
- high parasitemia in myofibrils (amastigotes)

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- 5 Chagas disease: disease stages
- 6 Chagas disease: disease stages
- 7 Chagas disease: disease stages
- 8 Chagas disease: disease stages

Chagas disease: disease stages

Chronic Phase

= Cardiomyopathy (cardiomegaly)



= Megaviscerae (esophagus, colon)



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Chagas disease: diagnosis

Parasite detection

- direct examination
- stained blood smears
- inoculation into mice
- in vitro culture
- xenodiagnosis
- PCR

Serological tests

- hemagglutination
- immunofluorescence
- ELISA
- complement fixation

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Chagas disease: treatment

Acute stage

= nifurtimox (8-16 mg/kg/day, 60-90 days)

= benzidazole (5-7 mg/kg/day, 30-120 days)

Chronic stage

= treat symptoms

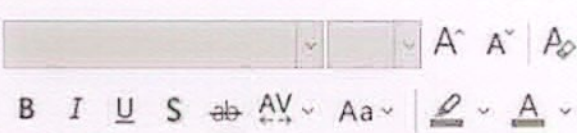
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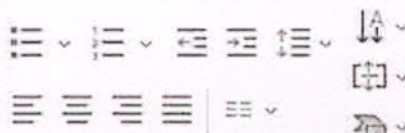
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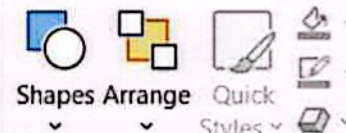
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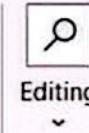
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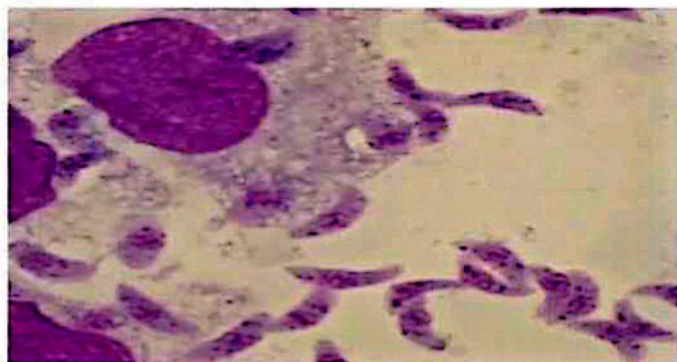
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Etiology

Toxoplasma gondii

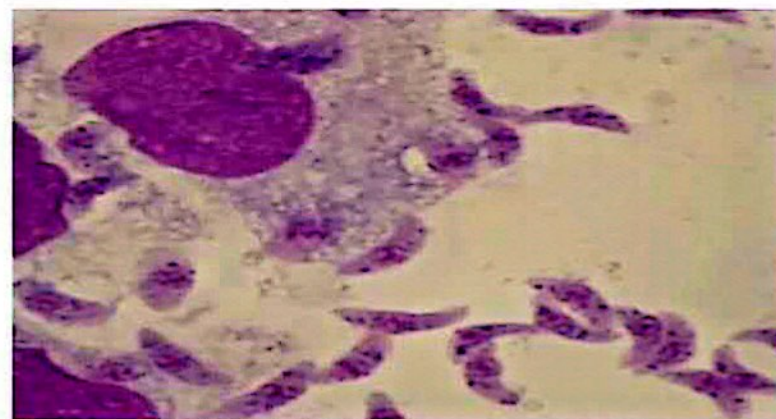
- An obligate intracellular parasite of the coccidia subclass displaying typical sexual and asexual life cycles



Etiology

Toxoplasma gondii

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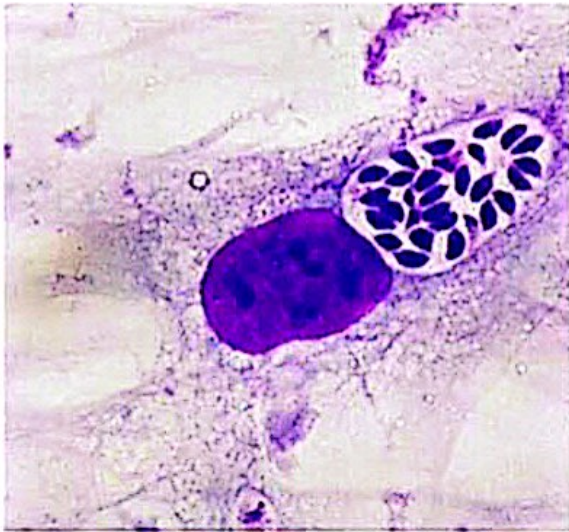
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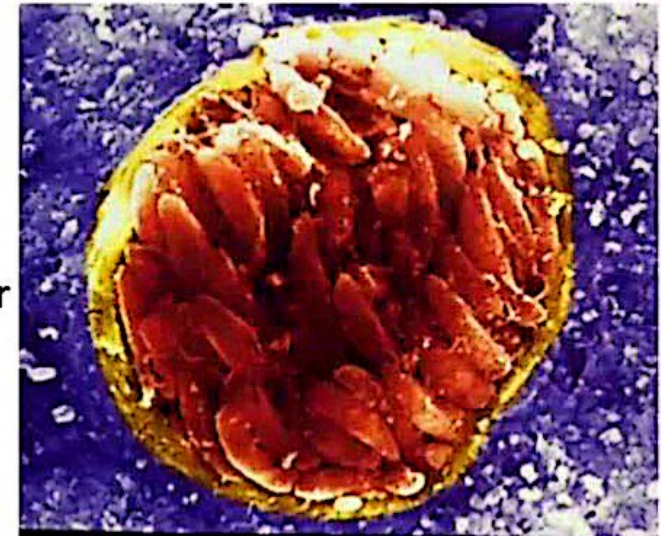
- There are three forms of *Toxoplasma gondii*



➤ There are three forms of *Toxoplasma gondii*



Tachyzoite – Rapidly reproducing form



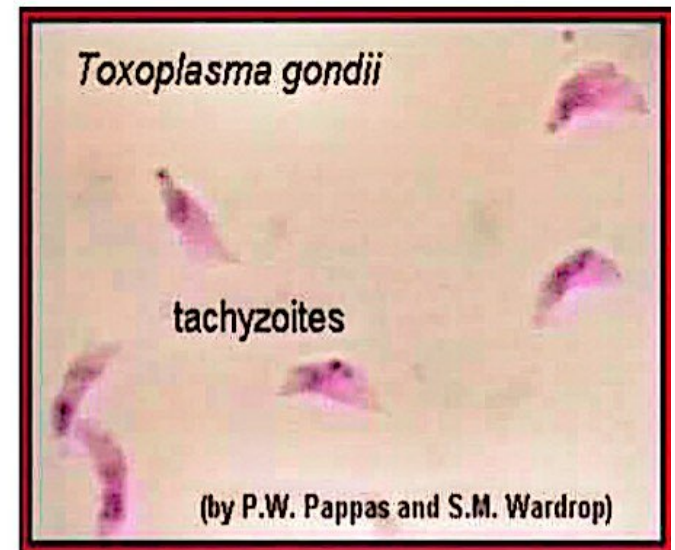
Bradyzoite – A slower reproducing form, contained in tissue



Sporozoite – Contained in oocysts

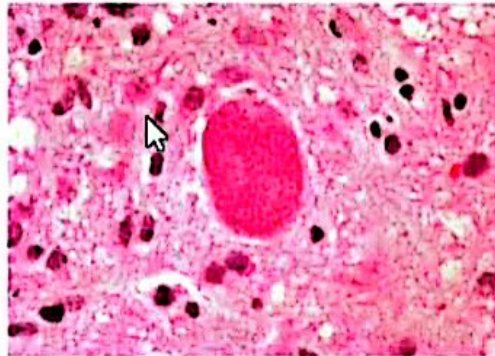
Tachyzoite

- Tachyzoites invade cells in the body in order to multiply. They invade a cell, multiply themselves, destroy the cell, and this releases more tachyzoites to infect other cells.
- Tachyzoites are seen in many tissues and organs throughout an infected body during the acute phase of the disease.
- The **acute phase** is also called **the extraintestinal phase** because it affects cells outside the intestines.
- Only cats show the intestinal phase of the infection.



Bradyzoites

- Two or three weeks after the first infection, the *Toxoplasma* microorganism begins to divide more slowly and a protective membrane forms around the parasite cells.
- The cysts containing the parasite cells are called zoitocysts and the cells inside the cysts are called bradyzoites.
- The tissue cysts are formed primarily in brain, eye, heart muscle, and skeletal muscle.
- Bradyzoites persist in tissue for many years, possibly for the life of the host.



Oocysts

- In cats, the *Toxoplasma* parasites infect the lining of the small intestine where they reproduce asexually.
- After a few days of rapid reproduction the cells transform into a sexual form, combine, and become enclosed in a cyst called an oocyst.
- Oocysts contain the sporozoite form of the *Toxoplasma* parasite.
- Oocysts are found in both wild and domestic cats but not in any other animals or birds.
- Animals become infected by ingesting food contaminated with oocysts from feces or with tissue cysts (bradyzoite) from the flesh of infected animals.
- Infected cats shed millions of oocysts in their feces during the two-week period of the intestinal phase of the infection
- These oocysts can survive in soil for several months and are the major source of infection.
- Once swallowed, the oocyst burst in the intestines and spreads to the rest of the body through the bloodstream.



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